

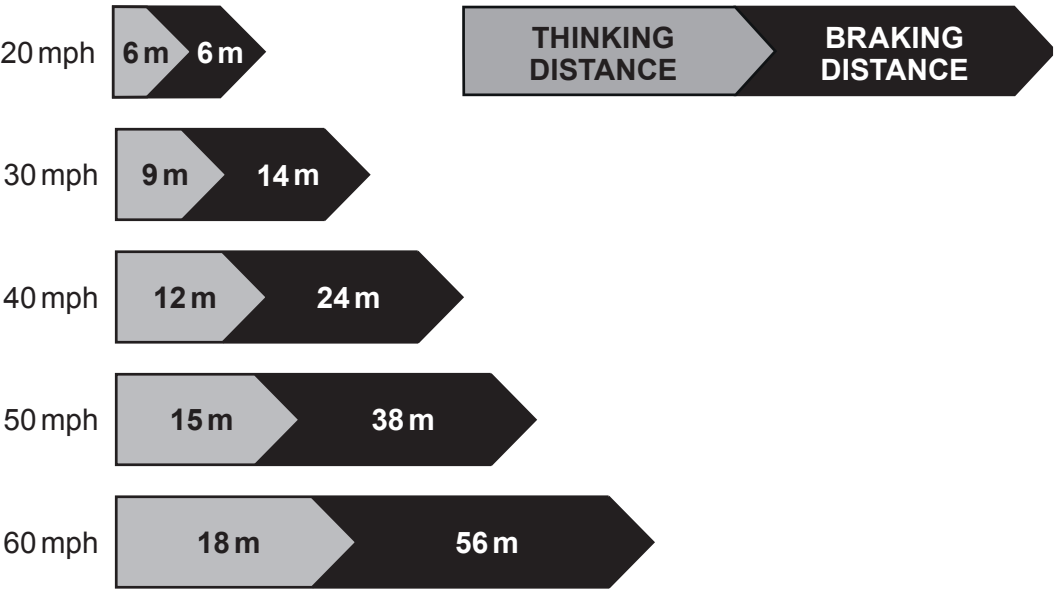
Answer **all** questions.

1. (a) When a car stops the overall stopping distance is made up of two distances: the thinking distance and the braking distance.
Increasing speed increases both the thinking distance and the braking distance.

(i) State **one** factor, other than speed, which increases the thinking distance. [1]

(ii) State **one** factor, other than speed, which increases the braking distance. [1]

(b) The diagram gives information about stopping distances at different speeds.



On a dangerous road, it is proposed to reduce the speed limit from 40 mph to 20 mph.

Bethan makes the following 3 suggestions.

- 1. The thinking distance will halve.
- 2. The braking distance will halve.
- 3. The overall stopping distance will halve.

Explain whether you agree with each suggestion.
Include data from the diagram to support your answer.

[3]

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